Scottish Journal of Political Economy

SCOTTISH ECONOMIC SOCIETY

ORIGINAL ARTICLE OPEN ACCESS

Conditional Gains: When AI Investment Enhances Firm Efficiency

Pantelis Kazakis

Adam Smith Business School, University of Glasgow, Glasgow, UK

Correspondence: Pantelis Kazakis (pantelis.kazakis@glasgow.ac.uk)

Received: 1 December 2025 | **Revised:** 29 January 2026 | **Accepted:** 3 March 2026

Keywords: artificial intelligence | financing horizon | firm efficiency | investor horizon | managerial ability | market power

ABSTRACT

The rapid adoption of artificial intelligence (AI) has raised questions about its effect on firm performance. Using a labor-based measure of AI investment, the baseline results show no direct association between AI investment and firm efficiency. The heterogeneity analysis indicates that efficiency gains materialize primarily when firms pair AI with capable managers, face stronger competitive pressure, have more stable institutional investor ownership, and are able to secure long-term debt.

JEL Classification: D40, E22, G30, H26, L11

1 | Introduction

The integration of artificial intelligence (AI) has become critically important for modern corporations, offering the potential to transform decision-making, streamline operations, and drive innovation (Babina et al. 2024). Yet, a key question remains: in which specific areas can AI truly deliver value and enhance firm performance?

Firm efficiency is of paramount importance for a corporation, as it directly impacts profitability, competitiveness, and long-term sustainability. Efficient firms utilize their resources effectively, reduce unnecessary costs, and maximize output (Baik et al. 2013). High efficiency also improves the firm's outcomes, enabling it to respond swiftly to shifts in consumer preferences, technological advancements, and competitive pressures (Ebben and Johnson 2005). Moreover, efficiency is closely linked to financial health, as efficient resource allocation can strengthen financial performance, improve returns to shareholders, and facilitate sustainable growth (Barney 1991).

In this study, I find that AI investment, by itself, does not show a direct association with firm efficiency. This suggests that simply increasing the share of AI-skilled workers in a firm does not automatically translate into measurable productivity gains as complementary investments might be needed (see also Brynjolfsson and Hitt 2000; Brynjolfsson et al. 2021). However, the analysis reveals conditions under which AI investment appears to be more beneficial. First, firms with lower market power tend to benefit more from AI adoption. A plausible explanation is that these firms operate in more competitive environments, where efficiency gains are essential for survival and growth (Syverson 2011). Facing tighter margins and less pricing power, such firms may be more incentivized to use AI strategically to maintain or improve their competitive position (Bloom et al. 2016; Porter 1980). In contrast, firms with greater market power may lack the same urgency to exploit AI for efficiency, as their dominant position already allows for operational slack.

Second, the evidence indicates that managerial ability is a key complement to AI adoption: firms led by higher-ability managers are better able to translate AI-related investments into operating improvements. This is consistent with the view that AI is not a plug-and-play technology but an organizational capability that requires effective leadership to deploy, integrate, and scale. High-ability managers are more likely to select appropriate use cases, redesign workflows, coordinate across functions (IT, operations, and business units), and invest in the complementary assets (i.e., data infrastructure, employee training, and change management) needed for AI to deliver measurable gains. In contrast, lower-ability firms may face implementation frictions and misallocation, so AI spending yields weaker or even adverse performance effects.

Third, the results show that the interaction between AI intensity and institutional investor turnover is negative, indicating that the performance implications of AI adoption depend on the horizon of the firm's investor base. Firms with higher portfolio turnover appear less able to convert AI-related investments into efficiency gains. This pattern is consistent with AI requiring time, experimentation, and complementary organizational changes before benefits materialize; under short-horizon ownership, managers may face stronger pressure to deliver immediate results, which can discourage the longer-term investments and learning needed for successful AI implementation. As a result, AI adoption in firms with high investor turnover is more likely to be associated with weaker operating improvements, aligning with a "short-termism" mechanism rather than a purely technological one.

Fourth, I find that the interaction between AI intensity and long-term debt is positive, suggesting that AI-related investments are more likely to translate into operating improvements when firms have greater access to stable, long-horizon financing. Because implementing AI typically involves sizable upfront costs (e.g., data infrastructure, software and cloud services, specialized talent, *inter alia*) firms may need patient capital to absorb adjustment frictions and allow benefits to materialize over time. Long-term debt can ease near-term funding constraints and reduce refinancing pressure, enabling managers to commit to multi-year AI projects and complementary investments such as training and process re-engineering. As a result, firms with higher long-term leverage appear better positioned to realize efficiency gains from AI adoption, consistent with financing capacity acting as a complement to technological transformation.

Why might AI investment not show a direct link to firm efficiency? Several factors can explain this outcome. AI investments often require substantial upfront costs and yield returns only after extended periods; thus, firms might not yet have experienced measurable efficiency gains within the study's period (Davenport et al. 2021). Additionally, AI resources could be directed toward strategic objectives unrelated directly to immediate operational efficiency (Cockburn et al. 2018). Firms also frequently face challenges integrating AI effectively into their existing frameworks (Forbes 2023). In addition, inadequate training or poor strategic choices might further impede the realization of efficiency benefits (Davenport and Rajeev 2018). Sectoral heterogeneity and firm-specific differences may also hide underlying relationships, as certain industries or individual firms could benefit significantly from AI while others may not (Heyman et al. 2021). Lastly, it is possible that efficiency gains become evident only after surpassing a certain level of AI adoption (Brynjolfsson et al. 2019).

This paper offers several contributions to the existing literature. First, it enriches the broad body of research on firm efficiency by showing that AI investment, when considered in isolation, does not necessarily lead to improved efficiency outcomes. This challenges the prevailing assumption that the adoption of cutting-edge technologies such as AI automatically translates into performance gains, and instead suggests that complementary factors (such as organizational readiness) may be critical in realizing AI's potential. Second, the findings offer a timely reflection on the current stage of the AI revolution in the corporate sector. The absence of a strong direct link between AI investment and efficiency indicates that the effects of AI may take longer to materialize. As such, this paper provides empirical support for the view that we are still in the early phases of technological diffusion, where firms are experimenting with AI capabilities but have not yet fully integrated them.

2 | Theoretical Considerations and Hypotheses

2.1 | Why the Average Effect of AI on Efficiency Can Be Nil or Very Small

One might expect AI adoption to raise productivity multiple times because it can change the production process itself rather than merely improving it at the margin. Technological improvements can automate routine cognitive tasks, speed up prediction and decision-making, and reduce error rates in forecasting, quality control, and logistics. This allows firms to produce the same output with fewer inputs or higher output with the same inputs (Agrawal et al. 2022; Autor et al. 2003). It can also enable better matching and allocation, as well as better scheduling in real time. In addition, advanced technologies such as AI can expand the feasible set of products and processes by uncovering patterns in large datasets, accelerating R&D, and enabling faster iteration in design and testing (Jordan and Mitchell 2015; LeCun et al. 2015). In these ways, AI functions as a general-purpose technology that complements both capital and skilled labor, with potentially large gains when firms operate under optimal environments.

In this section I argue that the efficiency gains from investing in artificial intelligence (AI) are inherently uneven across firms and are based on the economic and organizational environment in which a firm operates. The central premise is that AI is not a *panacea*: hiring AI-skilled workers does not, by itself, deliver large productivity or efficiency improvements. To convert AI-related inputs into measurable performance gains, firms must also build complementary assets that embed AI into day-to-day decisions (Czarnitzki et al. 2023). These complements are costly, take time to develop, and typically require coordination across multiple units. As a result, the average link between AI investment and efficiency can be modest or even non-present.

There are three main reasons why AI investment may not be associated with higher efficiency. First, AI requires substantial fixed costs and learning. As it is typical with any new technology, firms pass through an experimentation period whereby the use of the new technology generates noisy outcomes. During this process, productivity may decline as new processes are built and employees are retrained. Second, AI benefits may be present only under certain conditions. For example, better equipment, better organization and management, and highly-skilled workers, among others. Third, any positive effects of AI might require time until they translate as actual firm performance.

Since the effect of AI investment on firm efficiency is theoretically ambiguous, I present my first hypothesis as a null:

Hypothesis 1. *(Baseline effect of AI). AI investment has no effect on firm efficiency.*

2.2 | The Role of Market Structure

Market structure is central in shaping firms' incentives and the degree of discipline they face. In competitive markets, firms are under constant pressure to cut costs, streamline operations, and adapt quickly to rivals' moves, leaving less room for managerial slack (Nickell 1996; Tirole 1988). Mistakes or delayed responses are punished: managers who fail to react to emerging threats can quickly see their firm's position erode. By the same logic, when competition is intense, firms that invest in AI but fail to convert it into tangible operational improvements are more likely to cede market share to better-executing competitors (Brynjolfsson and McElheran 2016; Syverson 2004, 2011).

By contrast, firms with substantial market power can face weaker pressure to undertake disruptive organizational change (Giroud and Mueller 2010). Market power can create slack that allows firms to postpone costly restructuring, tolerate inefficiencies, or allocate AI resources to projects with strategic but non-efficiency goals (e.g., customer retention). As a result, AI investment may have a weaker association with measured operating efficiency in less competitive settings.

With the above in mind, I formulate the following hypothesis:

Hypothesis 2. *(Market structure). Higher markups negatively moderate the relationship between AI investment and firm efficiency. The efficiency gains from AI investment are greater in more competitive industries.*

2.3 | The Role of Managerial Ability

Even with similar external conditions, firms differ in their ability to convert AI inputs into operational outcomes. Implementing AI requires selecting use cases with clear business value, aligning incentives across units, integrating new tools into existing workflows, and managing change. These tasks are managerial in nature. Higher-ability managers are better positioned to evaluate which AI applications are economically meaningful, allocate talent to the highest-return projects, and overcome internal resistance to process redesign (Alekseeva et al. 2021; Gulati et al. 2025).

Prior research shows that managers can create or destroy value depending on the ability they possess. Managerial ability has been linked to a range of corporate outcomes, including innovation (Chen et al. 2015), earnings quality (Demerjian et al. 2013), and labor investment (Anderson et al. 2025). Building on this evidence, I argue that AI is unlikely to be a plug-and-play input. Lower-ability managers may still hire AI-skilled workers, but they may struggle to coordinate the complementary investments required for AI to be productive or to embed AI tools into routine decision-making. In that setting, AI talent can remain underutilized and the firm may not achieve measurable efficiency gains. Accordingly, I expect managerial ability to strengthen the efficiency return to AI adoption.

That said, I present the third hypothesis below:

Hypothesis 3. *(Managerial ability). Managerial ability positively moderates the relationship between AI investment and firm efficiency. The efficiency gains from AI investment are greater for firms with higher-ability managers.*

2.4 | The Implications of Short-Termism Among Institutional Investors

Investor horizon is an important determinant of firm outcomes. Prior work shows that firms with institutional investors with longer horizon have a lower equity cost (Attig et al. 2013). This result highlights the potential monitoring role of institutional investors. Furthermore, acquiring firms with more short-horizon institutional investors are more likely to lose value in the acquisition process (Gaspar et al. 2005). However, the broader effects of investor horizon are not one-sided. Some studies find that short-term-oriented ownership is associated with higher firm value (Döring et al. 2021), consistent with the idea that the threat of exit can discipline managers and reduce inefficiencies. Other evidence points in the opposite direction. An increase in short-term institutional ownership can reduce long-term investment (Cremers et al. 2020) while boosting short-run earnings, as managers may prioritize actions that satisfy short-term performance expectations at the expense of longer-horizon value creation.

Specifically, for the case of AI, early stages can produce volatile performance (Bresnahan and Trajtenberg 1995; Brynjolfsson et al. 2021; Bushee 1998; Edmans et al. 2022) as firms test models and reconfigure processes. As a result, when ownership is dominated by short-horizon investors (i.e., higher churn ratio), managers may face stronger pressure to prioritize short-term earnings stability. This pressure can discourage additional spending that is necessary for success (e.g., data infrastructure, employee training, and longer pilot periods) and can lead firms to terminate AI initiatives prematurely.

Conversely, a longer-horizon investor base can provide patience during the experimentation phase and can reward disciplined long-run value creation rather than short-run smoothness (Aghion et al. 2013; Appel et al. 2016). Under this view, short-horizon ownership weakens the link between AI investment and efficiency.

I present the fourth hypothesis as follows:

Hypothesis 4. *(Investor horizon). The churn rate of institutional investors negatively moderates the relationship between*

AI investment and firm efficiency. The positive effect of AI investment is attenuated in firms with higher investor turnover.

2.5 | Financing Horizon: Long-Term Debt and the Capacity to Absorb Adoption Costs

AI adoption typically involves multi-year investment programs, and many of the associated costs are intangible and hard to redeploy, such as data infrastructure, software and cloud services, and specialized human capital. The benefits of these investments may take time to materialize because implementation often requires experimentation, employee training, and changes to internal processes Brynjolfsson et al. (2021). During this adjustment period, firms can experience temporary disruptions that depress measured performance. Long-term debt can support AI implementation by providing more stable financing and reducing near-term refinancing pressure (Duchin et al. 2017; Myers 1977). With a longer financing horizon, managers are better able to maintain complementary spending on training, data projects, and process redesign, even when short-run conditions deteriorate. As a result, firms with greater long-term debt capacity may be more likely to complete the implementation cycle and realize efficiency gains from AI adoption.

My hypothesis is that long-term debt strengthens the link between AI adoption and efficiency, as it provides a more stable funding base and reduces short-term refinancing pressure. This stability can help firms sustain the complementary investments and internal adjustments needed for AI to generate operational improvements, such as training, data infrastructure, and process redesign. In this sense, long-term debt acts as a facilitator of the transition required to realize the potential efficiency gains from AI, rather than as an independent driver of efficiency.

Formally, I hypothesize that:

Hypothesis 5. *(Financing horizon). A firm's level of long-term debt positively moderates the relationship between AI investment and firm efficiency. Specifically, the positive effect of AI investment on efficiency is stronger for firms with greater long-term debt.*

3 | Methodology

3.1 | Datasets Used and Sample Selection

I rely on multiple data sources to address the research question. Firm-level accounting data are drawn from Compustat, while firm efficiency is measured using the Data Envelopment Analysis (DEA) approach proposed by Demerjian et al. (2012). AI investment is based on the share of AI-related workers in a firm, as constructed by Babina et al. (2024). Specifically, I utilize the AI measure they offer, which is derived from the Cognism data based on resumes and extends back to before the year 2000. Firms operating in the financial and utilities sectors are omitted from the analysis (SIC codes 4900–4999 and 6000–6999). All continuous variables are winsorized at the 1% level to mitigate the influence of outliers. After completing this process, the baseline sample comprises 51,685 firm-year observations over 1990–2018.

3.1.1 | Calculation of Firm Efficiency With DEA

Demerjian et al. (2012) utilize the DEA method to derive firm efficiency. DEA offers two main advantages. First, it ranks efficiency relative to the Pareto frontier, unlike parametric methods that benchmark against the average, which can be skewed by underperformers. Second, DEA avoids imposing arbitrary weights on inputs and outputs, instead deriving them endogenously from the data.

To clarify the procedure, DEA efficiency can be expressed mathematically as the following ratio:

$$\frac{\sum_{i=1}^{s} u_i y_{ik}}{\sum_{j=1}^{m} v_j x_{jk}} \quad k = 1, 2, \ldots, n. \tag{1}$$

In Equation (1), s denotes the number of outputs, m the number of inputs, and n the decision-making units (DMUs), which in this context are firms. Demerjian et al. (2012) employ one output and seven inputs, all sourced from the Compustat database. Revenue serves as the sole output, while the inputs include: net property, plant, and equipment (PP&E), net operating leases, net R&D, purchased goodwill, other intangible assets, cost of inventory, and selling, general, and administrative expenses (SG&A). For each output and inputs there are weights assigned. These are denoted by u and v, respectively. By y and x we denote the quantities of outputs and inputs.

The steps to derive firm efficiency using the DEA technique are as follows:

1. DMUs are grouped—typically by industry—based on similarities in the relationship between inputs and outputs, ensuring that efficiency is assessed among comparable units.
2. The next step is to maximize Equation (1) for each DMU by changing the weights u and v.
3. The optimal weights obtained are applied to the respective output and input quantities, with the results summed across all outputs (numerator) and all inputs (denominator). This generates a ratio-based efficiency score for each DMU.
4. All efficiency scores are subsequently normalized by the highest score within the group, producing an ordinal ranking of DMUs based on relative efficiency. The most efficient units receive a score of one, indicating optimal performance.
5. By design, the weights u and v are non-negative, reflecting the assumption that all inputs and outputs contribute positively to the production process. Given that the input and output quantities are also non-negative, the DEA efficiency score is bounded below by zero.

The optimization problem to be solved is as follows:

$$max_v(\theta) = (Sales) \cdot \big(v_1 CoGS + v_2 SG\&A + v_3 PPE + \\ + v_4 OpsLease + v_5 R\&D + v_6 Goodwill + v_7 OtherIntan\big)^{-1} \quad (2)$$

In Equation (2), stock variables—such as *Net PP&E, Net Operating Leases, Net R&D, Purchased Goodwill, and Other Intangible Assets*—are measured at the start of year t, whereas flow variables like *Cost of Inventory* and *SG&A* are measured over the course of year t. Demerjian et al. (2012) estimate DEA efficiency separately within each Fama–French industry to ensure that firms being compared share similar business models and cost structures.

3.1.2 | The AI Investment Measure

Herein, I offer additional details on the AI investment measure employed in the analysis. Babina et al. (2024) develop a novel proxy for firms' AI investments by examining the intensity of hiring workers with AI-related skills. Their approach relies on employee résumés to estimate the stock of AI-specialized personnel at each firm for which data is available. To construct this measure, they draw on a large dataset containing over 500 million individual employment profiles obtained from Cognism, a platform that aggregates professional resume data.[1]

Their second data source is Burning Glass, which provides information on over 180 million U.S. job postings from 2010 to 2018. Burning Glass collects data from more than 40,000 online job boards and company websites, employing advanced parsing techniques to structure and aggregate the information into labor market analytics. The dataset is highly granular, including fields such as, job title, location, occupation, and employer name. Crucially, each job posting is tagged with a rich set of specific skills.[2]

Finally, Babina et al. (2024) construct a human capital–based measure of firm-level AI investment using résumé data from Cognism. The process begins by analyzing job postings to empirically identify the skills most closely associated with AI. These AI-related skills are then matched within the résumé data to pinpoint AI-skilled workers. Finally, they compute the share of AI-skilled employees at each firm to derive a firm-level measure of AI investment.

3.2 | Econometric Model

To examine the association between AI investment and firm efficiency I use the following OLS model:

$$Efficiency_{i,t+1} = \beta_0 + \beta_1 AI_{i,t} + \beta_2 \mathbf{Controls}_{i,t} + \phi_i + \chi_t + \varepsilon_{i,t}. \quad (3)$$

In the above specification, *Efficiency* indicates the efficiency of a firm. *AI* denotes the share of *AI* workers in a firm, **Controls** is a vector of firm specific characteristics, while ϕ and χ represent firm- and year-fixed effects. Finally, ε is the error term. The dependent variable is measured 1 year ahead to allow sufficient time for AI-related investments to translate into observable improvements in firm efficiency. Firm fixed effects (ϕ_i) absorb all time-invariant differences across firms (e.g., business

TABLE 1 | Variable definitions.

Variable	Description
Efficiency	The total firm efficiency measure provided by Demerjian et al. (2012) and based on data envelopment analysis
AI	AI human capital investment is measured as the number of AI-related employees in a given year, scaled by the firm's total number of employees. Source: Babina et al. (2024)
Size	Firm assets (AT) in logs. Source: Compustat
RD	Research and development expenditures (XRD) in logs. Source: Compustat
CAPX	Ratio of capital expenditures ($CAPX$) to gross property, plant, and equipment ($PPEGT$). Source: Compustat
Leverage	Firm leverage. This is calculated from Compustat as $(DLC + DLTT)/AT$
INTAN	Intangibles (INT) scaled by total assets (AT). Source: Compustat
ROA	Earnings before interest and taxes divided by total assets (Compustat: $EBIT/AT$).
MTB	Market-to-book ratio (Compustat: $CSHO \times PRCC\_F/CEQ$)
Age	Firms' age in logs. Measured as the number of years since the firm first appears in the Compustat database, including the current year. Calculated by subtracting the initial year of observation from the current year and adding one
Sales_growth	Based on Compustat's variable SALE, calculated as: $(SALE_t - SALE_{t-1})/SALE_{t-1}$
Markup	The market power index calculated as in De Loecker et al. (2020). This variable is in logs
MA score rank	Managerial ability score decile ranks (by industry and year) provided by Demerjian et al. (2012)
Churn rate	Institutional investor churn rates as in Gaspar et al. (2005). Source: Thomson Reuters S34
LTB	Long-term debt calculated as $dltt/at$. Source: Compustat
Asset turnover	Asset turnover calculated as $sale/at$. Source: Compustat

models, industry niches, organizational culture, baseline technology, or persistent managerial quality) that may jointly affect AI adoption and efficiency. Year fixed effects (χ_t) control for economy-wide shocks and common trends (e.g., business cycles, regulatory changes, and broad advances in digital technologies) that influence both AI investment and firm efficiency in a given year. Standard errors are clustered at the firm level because observations are repeated within firms over time, so regression residuals may exhibit within-firm serial correlation and heteroskedasticity. Thus, by clustering at the firm-level our inference is robust to arbitrary correlation of errors within a firm across time.

I interpret the estimates as conditional associations, recognizing that adoption choices and unobserved time-varying shocks may still affect inference.

I include a standard set of firm characteristics that prior work links to both technology adoption and operating performance. Firm size (*Size*) controls for scale and resource capacity, which can facilitate the hiring of AI talent and the absorption of fixed costs. Research and development intensity (*RD*) and capital expenditures (*CAPX*) capture investment in intangible and tangible inputs that are complementary to digital technologies and may independently affect efficiency. Leverage (*Leverage*) proxies for financial constraints and risk-taking incentives that shape the feasibility and timing of organizational change. Intangible assets (*INTAN*) reflect the firm's knowledge base and organizational capital, which can influence the ability to redeploy data and analytics. Profitability (*ROA*) controls for internal funding and baseline operational performance, while growth opportunities (*MTB*) capture expectations about future expansion that may correlate with both AI investment and efficiency. Firm age (*Age*) accounts for life cycle and organizational rigidity, and sales growth (*Sales_growth*) controls for demand conditions and expansion dynamics that mechanically affect measured efficiency. Together, these controls help isolate the association between AI investment and subsequent efficiency from confounding variation in firms' investment policies, financial conditions, and growth prospects.

Table 1 shows the definitions of these variables.

4 | Results

Summary statistics are reported in Table 2. The average (median) firm efficiency is approximately 0.34 (0.29), suggesting that many firms operate with substantial inefficiencies. The share of AI investment remains relatively low, aligning with the figures reported in Babina et al. (2024). This is expected, as the measure used here represents a simple ratio rather than being multiplied by 100, as done in Liu and Zhang (2025).[3] This implies that throughout most of the sample period, only a limited number of firms employed a significant proportion of AI-related workers.

Table 3 reports Pearson correlations. In Panel A, firm efficiency is positively correlated with AI intensity (0.12) and is strongly related to firm size (0.54), consistent with scale effects in production. Efficiency also co-moves positively with R&D (0.34) and profitability (ROA; 0.17), while it is only

TABLE 2 | Summary statistics.

Variable	Obs.	Mean	Std. Dev.	Min	p25	p50	p75	Max
Efficiency	51,685	0.339	0.182	0.000	0.228	0.286	0.394	1.000
AI	51,685	0.000	0.001	0.000	0.000	0.000	0.000	0.010
Size	51,685	6.169	2.294	0.034	4.512	6.085	7.755	12.092
RD	51,685	1.688	2.006	0.000	0.000	0.697	3.172	6.225
CAPX	51,685	0.135	0.121	0.000	0.060	0.099	0.167	0.953
Leverage	51,685	0.232	0.310	0.000	0.024	0.181	0.336	4.382
INTAN	51,685	0.154	0.185	0.000	0.005	0.077	0.244	0.752
ROA	51,685	−0.044	0.457	−11.407	−0.026	0.038	0.080	0.427
MTB	51,685	3.065	6.426	−37.294	1.230	2.121	3.708	48.028
Age	51,685	2.687	0.834	0.693	2.079	2.708	3.332	4.220
Sales_growth	51,685	0.220	0.804	−1.000	−0.018	0.085	0.234	9.049
Markup	44,465	0.403	0.463	−2.506	0.105	0.281	0.585	2.986
MA score rank	44,465	0.561	0.283	0.100	0.300	0.600	0.800	1.000
Churn ratio	21,750	0.258	0.123	0.000	0.190	0.232	0.284	0.876
LTB	44,486	0.171	0.190	0.000	0.003	0.123	0.270	0.951
Asset turnover	42,347	1.190	0.762	0.110	0.658	1.023	1.524	4.134

Note: The definitions of all variables can be found in Table 1.

TABLE 3 | Correlation matrix.

Panel A: Core variables

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
(1) Efficiency	1.00										
(2) AI	0.12	1.00									
(3) Size	0.54	0.09	1.00								
(4) RD	0.34	0.21	0.38	1.00							
(5) CAPX	0.03	0.04	−0.13	−0.01	1.00						
(6) Leverage	−0.02	−0.04	0.05	−0.10	−0.12	1.00					
(7) INTAN	0.01	0.07	0.22	0.10	−0.06	0.09	1.00				
(8) ROA	0.17	0.01	0.28	0.03	−0.02	−0.39	0.03	1.00			
(9) MTB	0.09	0.05	−0.01	0.09	0.12	−0.11	−0.01	0.04	1.00		
(10) Age	0.15	0.00	0.37	0.14	−0.38	0.05	0.08	0.12	−0.06	1.00	
(11) Sales_growth	−0.02	0.00	−0.11	−0.01	0.29	−0.03	0.00	−0.05	0.08	−0.21	1.00

Panel B: Additional variables vs. core controls

	Efficiency	AI	Size	RD	CAPX	Leverage	INTAN	ROA	MTB	Age	Sales_ growth
Markup	0.35	0.14	−0.02	0.28	0.16	−0.12	0.11	0.06	0.14	−0.13	0.05
MA score rank	0.43	0.07	−0.03	0.11	0.16	−0.07	−0.14	0.07	0.10	−0.03	0.06
Churn ratio	−0.05	−0.05	−0.16	−0.05	0.15	0.00	−0.09	−0.04	0.01	−0.21	0.12
LTB	0.01	−0.05	0.25	−0.09	−0.14	0.72	0.18	−0.06	−0.09	0.09	−0.04
Asset turnover	−0.04	−0.08	−0.22	−0.30	−0.07	0.03	−0.16	−0.02	−0.04	0.05	−0.07

Note: Correlations are Pearson correlations computed pairwise. Variable definitions appear in Table 1. Panel B reports correlations between auxiliary variables and the core controls used throughout; correlations among auxiliary variables are not shown because these variables are used in separate subsample specifications.

weakly related to leverage (−0.02) and capital expenditures (0.03). Among controls, several correlations align with standard patterns: leverage is negatively associated with ROA (−0.39), and age is negatively related to sales growth (−0.21) and CAPX (−0.38). Panel B shows that the auxiliary variables are meaningfully associated with the core controls. For example, markup correlates positively with efficiency (0.35) and AI (0.14), managerial ability (MA score rank) is positively related to efficiency (0.43), and asset turnover is negatively correlated with size (−0.22) and R&D (−0.30). Overall, the correlations do not suggest severe pairwise collinearity among the baseline regressors.

Turning to the baseline analysis in Table 4, the results indicate no statistically significant relationship between AI investment and firm efficiency in isolation. Across all model specifications, the coefficient for the AI variable is insignificant. To account for potential lagged effects (where AI adoption may require time to manifest), I extended the analysis to include future efficiency measures up to 5 years ahead. As shown in Table A1, these lagged specifications similarly reveal no significant association. To explore sectoral variation, I applied the baseline model to individual industries (Table A2). With the exception of *Construction*, where AI investment correlates negatively with efficiency, no industry exhibits a statistically meaningful link between AI investment and efficiency outcomes.

I next examine whether the relationship between AI investment and firm efficiency varies across competitive environments. As shown in Table 5, the interaction term indicates that the efficiency implications of AI depend on product-market power. In column (1), the coefficient on AI is positive but not statistically significant, whereas the interaction term between AI and markup is negative and significant, implying that the marginal association between AI and firm efficiency declines as markups rise. Put differently, AI is more strongly linked to efficiency improvements in low-markup (more competitive) settings, while this link weakens, and can turn adverse, for firms with substantial market power. Because both DEA-based efficiency and the markup measure are related to revenues and costs, one concern is that this interaction could partly reflect overlapping measurement components. To address this, I re-estimate the same heterogeneity specification using an alternative operating outcome that does not rely on the DEA construction. Column (2) provides complementary

TABLE 4 | Baseline results.

	(1)	(2)
AI	0.014	−0.602
	[0.013]	[−0.579]
Size		0.030***
		[13.310]
RD		−0.007***
		[−3.026]
CAPX		−0.005
		[−0.562]
Leverage		0.023***
		[4.960]
INTAN		−0.154***
		[−15.163]
ROA		0.020***
		[6.858]
MTB		0.001***
		[6.487]
Age		−0.000
		[−0.084]
Sales_growth		0.004***
		[3.050]
Constant	0.340***	0.179***
	[1016.773]	[10.919]
Observations	59,836	51,685
Adjusted R^2	0.633	0.654
Firm effects	Yes	Yes
Year effects	Yes	Yes

Note: The dependent variable in all regressions is firm efficiency. The number of observations drops from column (1) to column (2) due to missing observations in some of the control variables. T-statistics, shown in brackets, are robust and clustered at the firm level. Stars, ***, **, *, indicate statistical significance at the 1%, 5%, and 10%, respectively. The definitions of all variables can be found in Table 1.

TABLE 5 | AI, markups, and firm outcomes.

	Dep var.: Firm efficiency	Dep var.: Asset turnover
	(1)	(2)
AI	2.177	−9.055*
	[1.619]	[−1.764]
Markup	0.108***	−0.086***
	[21.296]	[−4.776]
AI × Markup	−3.259**	12.255***
	[−2.099]	[2.604]
Size	0.026***	−0.141***
	[14.087]	[−14.141]
RD	−0.009***	0.016**
	[−4.568]	[2.124]
CAPX	−0.019**	−0.181***
	[−2.439]	[−5.701]
Leverage	0.011***	0.122***
	[2.861]	[5.233]
INTAN	−0.141***	−0.454***
	[−16.486]	[−11.953]
ROA	0.005**	0.023
	[2.537]	[1.238]
MTB	0.001***	−0.000
	[5.300]	[−0.939]
Age	0.006	0.140***
	[1.602]	[9.153]
Sales_growth	0.005***	0.005
	[4.537]	[0.881]
Constant	0.148***	1.743***
	[10.558]	[29.836]
Observations	44,465	42,347
Adjusted R^2	0.696	0.823
Firm effects	Yes	Yes
Year effects	Yes	Yes

Note: T-statistics, shown in brackets, are robust and clustered at the firm level. Stars, ***, **, *, indicate statistical significance at the 1%, 5%, and 10%, respectively. The definitions of all variables can be found in Table 1.

evidence using asset turnover. Although AI is negatively related to turnover on average, the positive and significant interaction with markup indicates that this association attenuates and can reverse as markups increase. A plausible interpretation is that, in high-markup settings, AI is deployed toward revenue-side or customer-facing applications (e.g., sales and customer retention) rather than toward the more disruptive cost-side reorganization that would manifest as higher overall efficiency. Overall, the estimates are consistent with the view that competitive pressure strengthens firms' incentives to translate AI capability into measurable operational gains, whereas market power weakens the link between AI investment and operating efficiency.

Table 6 reports cross-sectional heterogeneity in the association between AI investment and firm efficiency along three

TABLE 6 | Cross-sectional heterogeneity: Managerial ability, churn, and LTB.

	(1)	(2)	(3)
AI	−4.212***	2.961	−1.668
	[−3.329]	[1.405]	[−1.520]
MA score rank	0.118***		
	[30.987]		
AI × MA score rank	6.069**		
	[2.397]		
Churn ratio		0.022*	
		[1.916]	
AI × Churn ratio		−10.334*	
		[−1.891]	
LTB			−0.010
			[−1.433]
AI × LTB			13.375**
			[2.086]
Size	0.029***	0.021***	0.026***
	[17.093]	[5.239]	[12.863]
RD	−0.008***	−0.003	−0.007***
	[−4.586]	[−0.981]	[−3.455]
CAPX	−0.037***	−0.033**	−0.010
	[−4.871]	[−2.405]	[−1.201]
Leverage	0.010***	0.014	
	[2.607]	[0.976]	
INTAN	−0.111***	−0.151***	−0.131***
	[−13.753]	[−9.669]	[−13.643]
ROA	0.008***	0.040***	0.013***
	[3.334]	[3.411]	[4.575]
MTB	0.001***	0.001***	0.001***
	[5.236]	[3.185]	[5.499]
Age	0.003	−0.025***	0.007*
	[0.775]	[−3.220]	[1.737]
Sales_growth	0.002***	0.010***	0.005***
	[2.136]	[3.284]	[4.372]
Constant	0.112***	0.306***	0.189***
	[8.440]	[10.141]	[12.989]
Observations	44,465	21,750	44,486
Adjusted R^2	0.701	0.700	0.679
Firm effects	Yes	Yes	Yes
Year effects	Yes	Yes	Yes

Note: T-statistics, shown in brackets, are robust and clustered at the firm level. Stars, ***, **, *, indicate statistical significance at the 1%, 5%, and 10%, respectively. The definitions of all variables can be found in Table 1.

dimensions: managerial ability, investor horizon, and long-term debt. In column (1), managerial ability is strongly positively related to efficiency (MA score rank = 0.118), and the interaction term $AI \times MA$ score rank is positive and significant (6.069), indicating that the marginal association between AI and efficiency increases with managerial ability. This pattern is consistent with the view that higher-ability managers are better able to implement the complementary organizational changes required for AI to translate into operational gains.

Column (2) shows that the efficiency implications of AI are weaker in environments where the churn rate of institutional investors is high. While the churn ratio is weakly positively associated with efficiency (0.022), the interaction $AI\times$ Churn ratio is negative and marginally significant (−10.334), suggesting that high institutional investor turnover makes ownership less stable and weakens monitoring and commitment, reducing firms' incentives and ability to integrate AI into routines and workflows. The results in Column (3) suggest that a larger share of long-term debt is associated with stronger efficiency gains from AI, consistent with longer-maturity financing easing short-run repayment pressure and giving managers more scope to undertake the complementary investments and workflow redesign needed to translate AI capability into operational improvements.

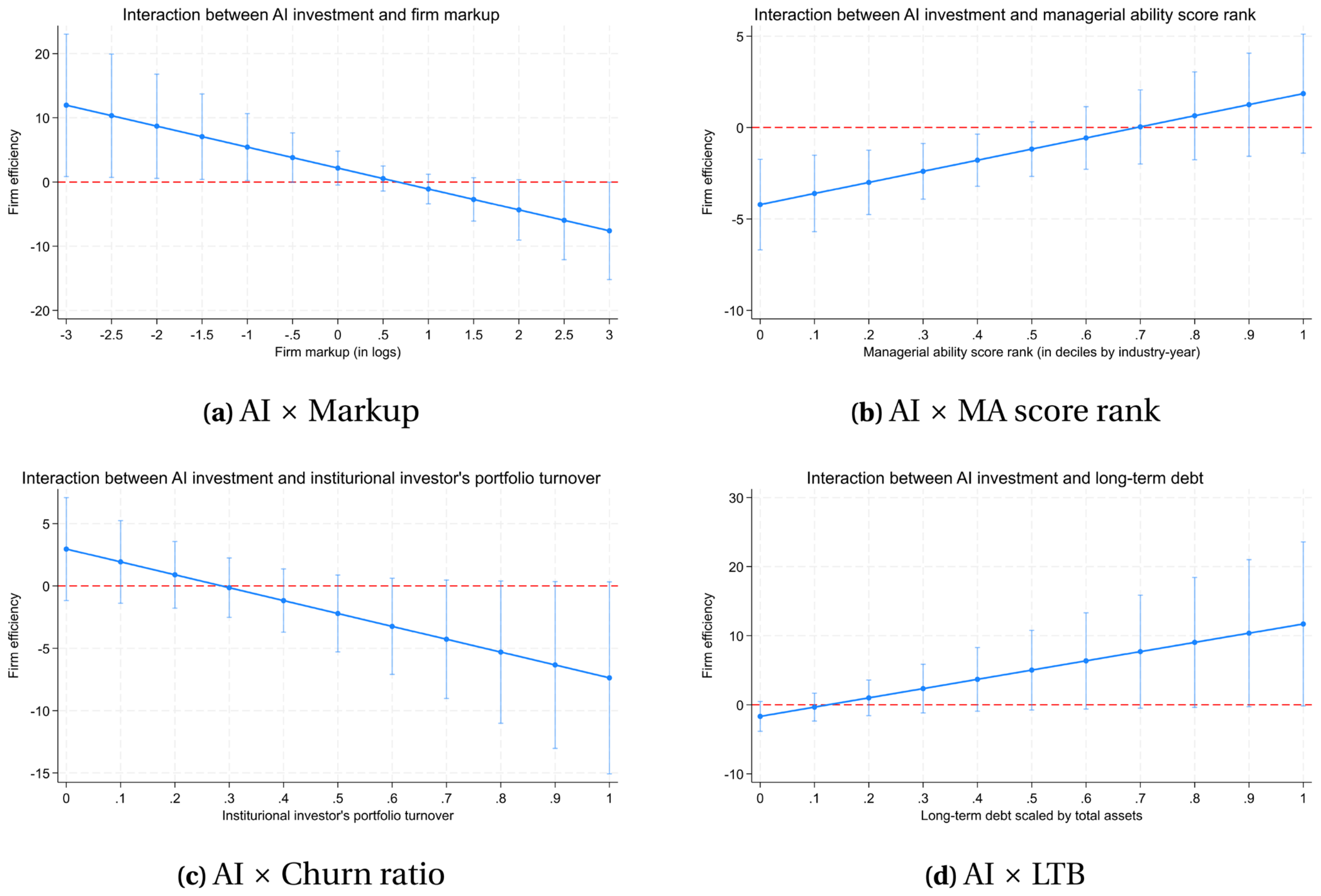
A characteristic of the AI investment variable is that it is highly skewed and contains many zeros, especially in earlier years. As a robustness test, I use log and inverse hyperbolic sine transformations that preserve information about relative intensity while reducing the influence of extreme values (Bellemare and Wichman 2020). This ensures that results are not driven by a small number of outliers and focuses the analysis on economically meaningful variation among adopting firms. I present the results of this analysis in the Table A3.

To better show the impact of the interaction terms, marginal effects are also displayed in Figure 1 below.

5 | Discussion, Implications, and Limitations

My findings imply that AI, apart from its tangible effects, is also understood as an organizational investment. For managers, the results highlight the importance of complementary investments. For AI to work, good managers should also invest in data infrastructure, governance, training, and workflow redesign. For investors and creditors, the evidence suggests that the returns to AI depend on the firm's incentives and horizons: competitive pressure and patient capital are more conducive to realizing efficiency gains from AI.

A limitation is that a labor-based AI measure (as the one used herein) captures capability and commitment, not actual model usage or the specific applications utilized inside the firm. This feature is common to many measures of emerging technologies, and it motivates my focus on heterogeneity: the same AI capability can produce different outcomes depending on market structure, managerial execution, and financing and ownership horizons.



(a) AI × Markup

(b) AI × MA score rank

(c) AI × Churn ratio

(d) AI × LTB

FIGURE 1 | Marginal effects.

6 | Conclusion

Understanding how investment in artificial intelligence (AI) is associated with firm efficiency is vital for both corporate strategy and economic policy. This study investigates this relationship and finds no direct association between AI investment and firm efficiency. I find, however, that the returns to AI are larger in more competitive settings and are amplified by higher managerial ability, while they are weakened when ownership is unstable (high institutional investor churn). Finally, firms with greater long-term debt exhibit a stronger AI–efficiency link, consistent with longer-maturity financing providing the slack needed to undertake complementary investments that allow AI to translate into measurable operational improvements. As AI becomes more widespread, future research using more updated data will be able to better assess AI's impact on firm performance, thus guiding both managerial decisions and targeted policy interventions.

Author Contributions

Pantelis Kazakis: conceptualization, investigation, writing – original draft, methodology, validation, visualization, writing – review and editing, software, formal analysis, data curation.

Endnotes

[1] For further details on the dataset, see Fedyk and Hodson (2023).

[2] For a comprehensive description of the dataset, see Hershbein and Kahn (2018).

[3] If multiplied by 100, the resulting statistics would be: mean = 0.03, standard deviation = 0.142, minimum = 0, 25th percentile = 0, median = 0, 75th percentile = 0, and maximum = 1.04.

References

Aghion, P., J. VAN Reenen, and L. Zingales. 2013. "Innovation and Institutional Ownership." *American Economic Review* 103, no. 1: 277–304.

Agrawal, A., J. Gans, and A. Goldfarb. 2022. *Prediction Machines, Updated and Expanded: The Simple Economics of Artificial Intelligence*. Harvard Business Press.

Alekseeva, L., J. Azar, M. Giné, S. Samila, and B. Taska. 2021. "The Demand for AI Skills in the Labor Market." *Labour Economics* 71 August 1, 2021: 102002.

Anderson, M., P. Sherer, and D. Yu. 2025. "Managerial Ability and Labor Investment." *Management Science* 71, no. 9: 8072–8095.

Appel, I. R., T. A. Gormley, and D. B. Keim. 2016. "Passive Investors, Not Passive Owners." *Journal of Financial Economics* 121, no. 1: 111–141.

Attig, N., S. Cleary, S. El Ghoul, and O. Guedhami. 2013. "Institutional Investment Horizons and the Cost of Equity Capital." *Financial Management* 42, no. 2: 441–477.

Autor, D. H., F. Levy, and R. J. Murnane. 2003. "The Skill Content of Recent Technological Change: An Empirical Exploration." *Quarterly Journal of Economics* 118, no. 4: 1279–1333.

Babina, T., A. Fedyk, A. He, and J. Hodson. 2024. "Artificial Intelligence, Firm Growth, and Product Innovation." *Journal of Financial Economics* 151: 103745.

Baik, B., J. Chae, S. Choi, and D. B. Farber. 2013. "Changes in Operational Efficiency and Firm Performance: A Frontier Analysis Approach." *Contemporary Accounting Research* 30, no. 3: 996–1026.

Barney, J. 1991. "Firm Resources and Sustained Competitive Advantage." *Journal of Management* 17, no. 1: 99–120.

Bellemare, M. F., and C. J. Wichman. 2020. "Elasticities and the Inverse Hyperbolic Sine Transformation." *Oxford Bulletin of Economics and Statistics* 82, no. 1: 50–61.

Bloom, N., M. Draca, and J. Van Reenen. 2016. "Trade Induced Technical Change? The Impact of Chinese Imports on Innovation, IT and Productivity." *Review of Economic Studies* 83, no. 1: 87–117.

Bresnahan, T. F., and M. Trajtenberg. 1995. "General Purpose Technologies 'Engines of Growth'?" *Journal of Econometrics* 65, no. 1: 83–108.

Brynjolfsson, E., and L. M. Hitt. 2000. "Beyond Computation: Information Technology, Organizational Transformation and Business Performance." *Journal of Economic Perspectives* 14, no. 4: 23–48.

Brynjolfsson, E., and K. McElheran. 2016. "The Rapid Adoption of Data-Driven Decision-Making." *American Economic Review* 106, no. 5: 133–139.

Brynjolfsson, E., D. Rock, and C. Syverson. 2019. "Artificial Intelligence and the Modern Productivity Paradox." In *The Economics of Artificial Intelligence: An Agenda 23*. University of Chicago Press.

Brynjolfsson, E., D. Rock, and C. Syverson. 2021. "The Productivity J-Curve: How Intangibles Complement General Purpose Technologies." *American Economic Journal: Macroeconomics* 13, no. 1: 333–372.

Bushee, B. J. 1998. "The Influence of Institutional Investors on Myopic R&D Investment Behavior." *Accounting Review* 73, no. 3: 305–333.

Chen, Y., E. J. Podolski, and M. Veeraraghavan. 2015. "Does Managerial Ability Facilitate Corporate Innovative Success?" *Journal of Empirical Finance* 34: 313–326.

Cockburn, I. M., R. Henderson, and S. Stern. 2018. "The Impact of Artificial Intelligence on Innovation: An Exploratory Analysis." In *The Economics of Artificial Intelligence: An Agenda*, 115–146. University of Chicago Press.

Cremers, M., A. Pareek, and Z. Sautner. 2020. "Short-Term Investors, Long-Term Investments, and Firm Value: Evidence From Russell 2000 Index Inclusions." *Management Science* 66, no. 10: 4535–4551.

Czarnitzki, D., G. P. Fernández, and C. Rammer. 2023. "Artificial Intelligence and Firm-Level Productivity." *Journal of Economic Behavior & Organization* 211: 188–205.

Davenport, T. H., A. Guha, and w. Dhruv. 2021. "How to Design an AI Marketing Strategy." Harvard Business Review, 108–116.

Davenport, T., and R. Rajeev. 2018. "Artificial Intelligence for the Real World." Harvard Business Review.

De Loecker, J., J. Eeckhout, and G. Unger. 2020. "The Rise of Market Power and the Macroeconomic Implications." *Quarterly Journal of Economics* 135, no. 2: 561–644.

Demerjian, P., B. Lev, and S. McVay. 2012. "Quantifying Managerial Ability: A New Measure and Validity Tests." *Management Science* 58, no. 7: 1229–1248.

Demerjian, P. R., B. Lev, M. F. Lewis, and S. E. McVay. 2013. "Managerial Ability and Earnings Quality." *Accounting Review* 88, no. 2: 463–498.

Döring, S., W. Drobetz, S. El Ghoul, O. Guedhami, and H. Schröder. 2021. "Institutional Investment Horizons and Firm Valuation Around the World." *Journal of International Business Studies* 52, no. 2: 212–244.

Duchin, R., T. Gilbert, J. Harford, and C. Hrdlicka. 2017. "Precautionary Savings With Risky Assets: When Cash Is Not Cash." *Journal of Finance* 72, no. 2: 793–852.

Ebben, J. J., and A. C. Johnson. 2005. "Efficiency, Flexibility, or Both? Evidence Linking Strategy to Performance in Small Firms." *Strategic Management Journal* 26, no. 13: 1249–1259.

Edmans, A., V. W. Fang, and A. H. Huang. 2022. "The Long-Term Consequences of Short-Term Incentives." *Journal of Accounting Research* 60, no. 3: 1007–1046.

Fedyk, A., and J. Hodson. 2023. "Trading on Talent: Human Capital and Firm Performance." *Review of Finance* 27, no. 5: 1659–1698.

Forbes. 2023. "10 Hurdles Companies are Facing with AI (and How to Overcome Them)." *Forbes*. https://www.forbes.com/councils/theyec/2023/10/25/10-hurdles-companies-are-facing-when-implementing-ai-and-how-to-overcome-them/.

Gaspar, J.-M., M. Massa, and P. Matos. 2005. "Shareholder Investment Horizons and the Market for Corporate Control." *Journal of Financial Economics* 76, no. 1: 135–165.

Giroud, X., and H. M. Mueller. 2010. "Does Corporate Governance Matter in Competitive Industries?" *Journal of Financial Economics* 95, no. 3: 312–331.

Gulati, P., A. Marchetti, P. Puranam, and V. Sevcenko. 2025. "Generative AI Adoption and Higher Order Skills." arXiv:2503.09212.

Hershbein, B., and L. B. Kahn. 2018. "Do Recessions Accelerate Routine-Biased Technological Change? Evidence From Vacancy Postings." *American Economic Review* 108, no. 7: 1737–1772.

Heyman, F., P.-J. Norbäck, and L. Persson. 2021. "Artificial Intelligence, Robotics, Work and Productivity: The Role of Firm Heterogeneity." Working Paper 1382. IFN Working Paper.

Jordan, M. I., and T. M. Mitchell. 2015. "Machine Learning: Trends, Perspectives, and Prospects." *Science* 349, no. 6245: 255–260.

LeCun, Y., Y. Bengio, and G. Hinton. 2015. "Deep Learning." *Nature* 521, no. 7553: 436–444.

Liu, Y., and Z. Zhang. 2025. "Unlocking Operational Efficiency: How AI Human Capital Investment Enhances Data Processing Efficiency?" *Economics Letters* 247: 112147.

Myers, S. C. 1977. "Determinants of Corporate Borrowing." *Journal of Financial Economics* 5, no. 2: 147–175.

Nickell, S. J. 1996. "Competition and Corporate Performance." *Journal of Political Economy* 104: 724–746.

Porter, M. 1980. *Competitive Strategy: Techniques for Analyzing Industries and Competitors*. Vol. 1. Free Press.

Syverson, C. 2004. "Market Structure and Productivity: A Concrete Example." *Journal of Political Economy* 112: 1181–1222.

Syverson, C. 2011. "What Determines Productivity?" *Journal of Economic Literature* 49, no. 2: 326–365.

Tirole, J. 1988. *The Theory of Industrial Organization*. MIT Press.

Appendix A

TABLE A1 | Baseline model with future values for the dependent variable.

	(1)	(2)	(3)	(4)
AI	−1.121	−1.711	−1.738	−2.133
	[−0.972]	[−1.499]	[−1.420]	[−1.626]
Size	0.023***	0.019***	0.016***	0.014***
	[10.000]	[7.963]	[6.600]	[5.623]
RD	−0.009***	−0.009***	−0.008***	−0.007***
	[−4.036]	[−4.097]	[−3.627]	[−2.949]
CAPX	−0.016*	−0.009	−0.012	−0.011
	[−1.720]	[−0.984]	[−1.435]	[−1.288]
Leverage	0.019***	0.018***	0.016***	0.015***
	[4.128]	[3.682]	[3.103]	[2.797]
INTAN	−0.133***	−0.118***	−0.106***	−0.090***
	[−13.188]	[−11.526]	[−9.536]	[−7.522]
ROA	0.008**	0.005*	0.005*	−0.000
	[2.516]	[1.882]	[1.919]	[−0.134]
MTB	0.001***	0.000***	0.000**	0.000***
	[4.427]	[2.919]	[2.379]	[2.918]
Age	0.005	0.006	0.007	0.005
	[1.077]	[1.217]	[1.558]	[1.032]
Sales_growth	0.002*	−0.002	0.001	0.000
	[1.798]	[−1.598]	[0.780]	[0.155]
Constant	0.217***	0.244***	0.257***	0.270***
	[12.791]	[13.770]	[14.279]	[14.398]
Observations	47,485	43,548	39,870	36,496
Adjusted R^2	0.656	0.662	0.666	0.669
Firm effects	Yes	Yes	Yes	Yes
Year effects	Yes	Yes	Yes	Yes

Note: In column (1), the dependent variable is firm efficiency measured at year $t+2$; in column (2), at $t+3$; in column (3), at $t+4$; and in column (4), at $t+5$. *T*-statistics are shown in brackets and are robust to heteroskedasticity with clustering at the firm level. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. The definitions of all variables can be found in thistable.

TABLE A2 | Baseline results by industry.

Variables	Mining (1)	Construction (2)	Manufacturing (3)	Transportation (4)	Wholesale (5)	Retail (6)
AI	7.318	−30.305***	1.445	−0.977	−1.326	6.024
	[1.052]	[−3.616]	[1.012]	[−0.238]	[−0.595]	[0.688]
Size	0.006	0.044***	0.033***	0.027***	0.069***	0.048***
	[0.702]	[2.796]	[8.140]	[3.008]	[5.213]	[6.142]
RD	0.010	0.017	0.000	−0.013	−0.001	−0.023
	[0.545]	[0.992]	[0.135]	[−1.255]	[−0.124]	[−1.389]
CAPX	0.042	0.080	0.013	−0.047	−0.028	−0.007
	[0.982]	[1.670]	[0.972]	[−1.350]	[−1.074]	[−0.198]
Leverage	0.027	0.009	0.030***	0.002	0.009	0.036**
	[1.348]	[0.449]	[3.721]	[0.164]	[0.244]	[1.989]
INTAN	−0.278***	−0.149*	−0.176***	−0.188***	−0.137**	−0.196***
	[−3.935]	[−1.902]	[−11.421]	[−4.626]	[−2.383]	[−4.486]
ROA	0.050**	−0.003	0.025***	0.006	0.027	0.044***
	[2.210]	[−0.448]	[4.845]	[0.754]	[1.184]	[3.229]
MTB	0.000	−0.000	0.001***	−0.000	0.000	0.000
	[0.782]	[−0.169]	[5.900]	[−0.287]	[0.326]	[1.113]
Age	0.000	−0.085***	0.011*	−0.036**	−0.017	−0.014
	[0.011]	[−2.999]	[1.936]	[−2.099]	[−1.056]	[−0.995]
Sales_growth	0.010**	0.019***	0.003	0.005	−0.001	0.000
	[2.533]	[2.994]	[1.407]	[0.883]	[−0.109]	[0.022]
Constant	0.292***	0.284**	0.115***	0.280***	0.014	0.096
	[4.939]	[2.232]	[4.506]	[3.863]	[0.178]	[1.616]
Observations	2660	762	26,887	3404	2095	3915
Adjusted R^2	0.534	0.663	0.672	0.623	0.708	0.712
Firm effects	Yes	Yes	Yes	Yes	Yes	Yes
Year effects	Yes	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable in all regressions is firm efficiency. T-statistics, shown in brackets, are robust and clustered at the firm level. Stars, ***, **, *, indicate statistical significance at the 1%, 5%, and 10%, respectively. The definitions of all variables can be found in Table A1.

TABLE A3 | AI, alternative AI measurement, and heterogeneity tests (firm efficiency).

	Baseline		Market power		MA score rank		Churn ratio		LTB	
	Log	IHS	Log	IHS	Log	IHS	Log	IHS	Log	IHS
AI measured in ⇒	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**	**(9)**	**(10)**
AI	−0.604	−0.602	2.188	2.177	−4.232***	−4.212***	2.978	2.961	−1.676	−1.668
	[−0.578]	[−0.579]	[1.621]	[1.619]	[−3.331]	[−3.329]	[1.407]	[1.405]	[−1.521]	[−1.520]
Markup			0.108***	0.108***						
			[21.297]	[21.296]						
AI × Markup			−3.278**	−3.259**						
			[−2.101]	[−2.099]						
MA score rank					0.118***	0.118***				
					[30.986]	[30.987]				
AI × MA score rank					6.099**	6.069**				
					[2.398]	[2.397]				
Churn ratio							0.022*	0.022*		
							[1.917]	[1.916]		
AI × Churn ratio							−10.399*	−10.334*		
							[−1.894]	[−1.891]		
LTB									−0.010	−0.010
									[−1.433]	[−1.433]
AI × LTB									14.344**	13.375**
									[2.086]	[2.086]
Size	0.030***	0.030***	0.026***	0.026***	0.029***	0.029***	0.021***	0.021***	0.026***	0.026***
	[13.310]	[13.310]	[14.087]	[14.087]	[17.093]	[17.093]	[5.238]	[5.239]	[12.863]	[12.863]
RD	−0.007***	−0.007***	−0.009***	−0.009***	−0.008***	−0.008***	−0.003	−0.003	−0.007***	−0.007***
	[−3.026]	[−3.026]	[−4.568]	[−4.568]	[−4.586]	[−4.586]	[−0.981]	[−0.981]	[−3.455]	[−3.455]
CAPX	−0.005	−0.005	−0.019**	−0.019**	−0.037***	−0.037***	−0.033**	−0.033**	−0.010	−0.010
	[−0.560]	[−0.562]	[−2.439]	[−2.439]	[−4.870]	[−4.871]	[−2.404]	[−2.405]	[−1.201]	[−1.201]
Leverage	0.023**	0.023**	0.011***	0.011***	0.010***	0.010***	0.014	0.014		
	[4.960]	[4.964]	[2.861]	[2.861]	[2.607]	[2.607]	[0.976]	[0.976]		
INTAN	−0.154***	−0.154***	−0.141***	−0.141***	−0.111***	−0.111***	−0.151***	−0.151***	−0.131***	−0.131***
	[−15.163]	[−15.163]	[−16.486]	[−16.486]	[−13.753]	[−13.753]	[−9.669]	[−9.669]	[−13.643]	[−13.643]
ROA	0.020***	0.020***	0.005**	0.005**	0.008***	0.008***	0.040***	0.040***	0.013***	0.013***
	[6.858]	[6.858]	[2.537]	[2.537]	[3.334]	[3.334]	[3.411]	[3.411]	[4.575]	[4.575]
MTB	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***
	[6.487]	[6.487]	[5.300]	[5.300]	[5.236]	[5.236]	[3.184]	[3.185]	[5.498]	[5.499]
Age	−0.000	−0.000	0.006	0.006	0.003	0.003	−0.025***	−0.025***	0.007*	0.007*
	[−0.004]	[−0.004]	[1.602]	[1.602]	[0.775]	[0.775]	[−3.220]	[−3.220]	[1.737]	[1.737]
Sales_growth	0.004***	0.004***	0.005***	0.005***	0.002***	0.002***	0.010***	0.010***	0.005***	0.005***
	[3.050]	[3.050]	[4.537]	[4.537]	[2.136]	[2.136]	[3.284]	[3.284]	[4.372]	[4.372]

(Continues)

TABLE A3 | (Continued)

AI measured in ⇒	Baseline		Market power		MA score rank		Churn ratio		LTB	
	Log	IHS	Log	IHS	Log	IHS	Log	IHS	Log	IHS
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Constant	0.179***	0.179***	0.148***	0.148***	0.112***	0.112***	0.306***	0.306***	0.189***	0.189***
	[10.919]	[10.919]	[10.558]	[10.558]	[8.440]	[8.440]	[10.141]	[10.141]	[12.989]	[12.989]
Observations	51,685	51,685	44,465	44,465	44,465	44,465	21,750	21,750	44,486	44,486
Adjusted R^2	0.654	0.654	0.696	0.696	0.701	0.701	0.700	0.700	0.679	0.679
Firm effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: The dependent variable is firm efficiency. Columns vary by (i) the measure of AI—where Log denotes measurement as $\log(1+x)$ and IHS denotes measurement as $\log\left(x+\sqrt{x^2+1}\right)$—and (ii) the heterogeneity proxy (Baseline, Market power, MA score rank, Churn ratio, LTB). Brackets report t-statistics. All specifications include firm and year fixed effects. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.